

Data Pre-Processing Report

McDonald's Retail Analytics System

1. Overview

The preprocessing phase was performed to clean, transform, and prepare the raw datasets for analysis and dashboard creation.

The complete preprocessing was done using **Power Query (M Language)** in Microsoft Excel.

The project uses two datasets:

1. **Menu Items Dataset** – contains product information
2. **Order Details Dataset** – contains transactional order records

The main goal of preprocessing was to:

- Clean inconsistent data
 - Handle missing values
 - Create analytical columns
 - Merge datasets
 - Prepare the data for KPI calculation and visualization
-

2. Data Sources

A. Menu Items Dataset

Column	Description
menu_item_id	Unique product ID
item_name	Product name
category	Product category
price	Product price

B. Order Details Dataset

Column	Description
order_details_id	Transaction row ID
order_id	Order identifier
order_date	Order date
order_time	Order time
item_id	Ordered product ID

3. Pre-Processing Steps

3.1 Importing Data

Both datasets were imported from a local folder using Power Query.

Files Used

- mcdonal_s menu items.xlsx
- order_details.csv

The folder connection method was used to make the project dynamic and scalable.

3.2 Cleaning Menu Items Dataset

The following preprocessing steps were applied to the Menu Items table:

Removed Unnecessary Columns

System-generated columns such as:

- Hidden
- Kind
- Item
- Name

were removed because they were not required for analysis.

Renamed Columns

Columns were renamed for better readability.

Old Name	New Name
Data.Column1	menu_item_id
Data.Column2	item_name
Data.Column3	category
Data.Column4	price

Removed Invalid Rows

- Top header duplication row was removed
 - Rows containing null menu_item_id values were filtered out
-

Changed Data Types

Column	Data Type
menu_item_id	Whole Number
category	Text
price	Decimal Number

This ensured accurate calculations and proper aggregation.

4. Cleaning Order Details Dataset

The Order Details dataset required multiple transformations.

4.1 Header Promotion

The first row was promoted as column headers for proper table structure.

4.2 Handling Missing Values

The item_id column contained "NULL" values.

Steps Performed

- Replaced "NULL" with actual null values
- Applied Fill Down transformation to maintain data continuity

This ensured proper relationship mapping between tables.

4.3 Data Type Conversion

Correct data types were assigned to all fields.

Column	Data Type
order_details_id	Whole Number
order_id	Whole Number
item_id	Whole Number
order_date	Date
order_time	Time

4.4 Date Formatting

The date column was converted using English-US locale formatting to avoid incorrect date interpretation.

5. Feature Engineering

Additional columns were created to improve business analysis.

5.1 Month Name

A new column was created from order_date to analyze monthly sales trends.

Example:

- January
 - February
 - March
-

5.2 Day Name

A weekday column was created for day-wise sales analysis.

Example:

- Monday
 - Tuesday
 - Wednesday
-

5.3 Day Type

Days were classified into:

- Weekday
- Weekend

Logic Used

- Saturday and Sunday → Weekend
- Remaining days → Weekday

Purpose:

- Compare weekday vs weekend sales behavior
-

5.4 Time Slot Classification

Orders were grouped into business time slots.

Time Range	Time Slot
Before 12 PM	MORNING
12 PM – 3 PM	NOON
3 PM – 5 PM	EVENING
After 5 PM	NIGHT

Purpose:

- Identify peak ordering periods
 - Analyze customer ordering behavior
-

6. Data Merging

The two datasets were merged using:

- item_id
- menu_item_id

Join Type

Left Outer Join

This merge allowed product price information to be added to transactional records.

7. Revenue Preparation

After merging the tables:

- Product price was extracted into the order dataset
- The column was renamed as revenue

Revenue Assumption

Each row represents one quantity sold.

Therefore:

$$\text{Revenue} = \text{Price} \times 1 \quad \text{\text{Revenue}} = \text{\text{Price}} \times 1$$

This enabled:

- Revenue analysis
- KPI creation

- Category-wise sales analysis
-

8. Data Quality Checks

The following checks were performed:

Missing Values

Verified and cleaned missing values in:

- item_id
 - menu_item_id
 - order_date
 - order_time
-

Duplicate Records

Checked for:

- Duplicate transactions
- Duplicate menu items

No major duplication issues were found.

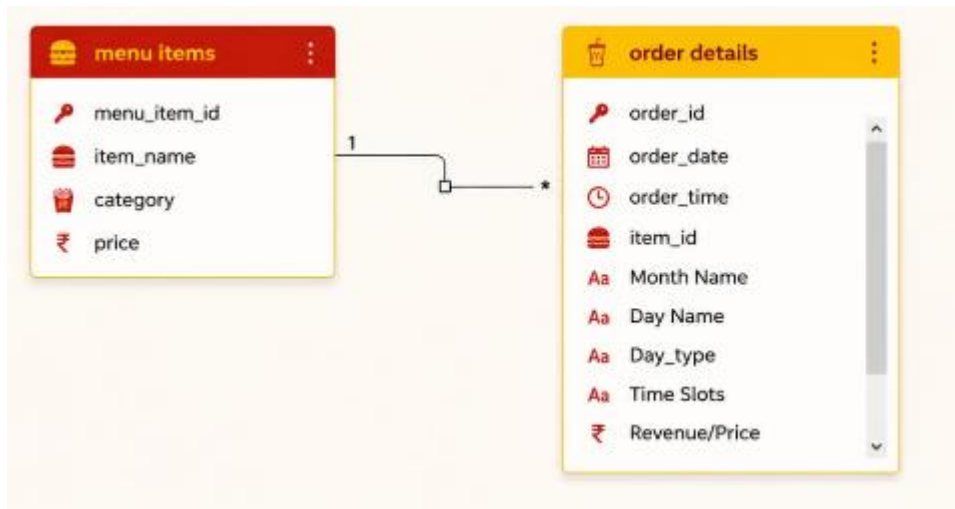
Relationship Validation

Validated the relationship between:

- item_id
- menu_item_id

to ensure accurate merging and calculations.

9. Final Dataset Structure



The final analytical dataset included:

Transaction Information

- order_id
- order_date
- order_time

Product Information

- item_name
- category
- price

Derived Columns

- Month Name
- Day Name
- Day_type
- Time Slot

Analytical Measure

- revenue

10. Dashboard Description

Two interactive dashboards were created to visualize business insights.

Dashboard 1 — Sales Dashboard

The Sales Dashboard focuses on revenue performance and product analysis.

KPIs Included

KPI	Description
Total Revenue	Total sales revenue generated
Total Orders	Total number of orders placed
Average Order Value	Average revenue per order
Most Ordered Item	Highest-selling product
Least Sold Item	Lowest-selling product



Visualizations Used

Top 5 Products by Orders

Displays the most frequently ordered products.

Revenue by Category

Shows category-wise revenue contribution.

Sales by Month

Analyzes monthly sales performance.

Sales by Day Type

Compares weekday vs weekend sales.

Key Facts from Sales Dashboard

- Burger category generated the highest revenue.
- Side Salad was the most ordered product.
- Chicken Snack Wrap was the least sold product.

- March generated the highest monthly sales.
- Weekday sales were higher than weekend sales.

Dashboard 2 — Order Pattern Dashboard

The Order Pattern Dashboard focuses on customer ordering behavior and operational trends.

KPIs Included

KPI	Description
Peak Hour	Time slot with maximum orders
Highest Sales Day	Day with highest sales
Highest Sales Month	Month with highest revenue
Peak Order Day Type	Highest ordering behavior

Visualizations Used

Sales by Day Time

Shows revenue distribution across:

- Morning
- Noon
- Evening
- Night

Sales by Month

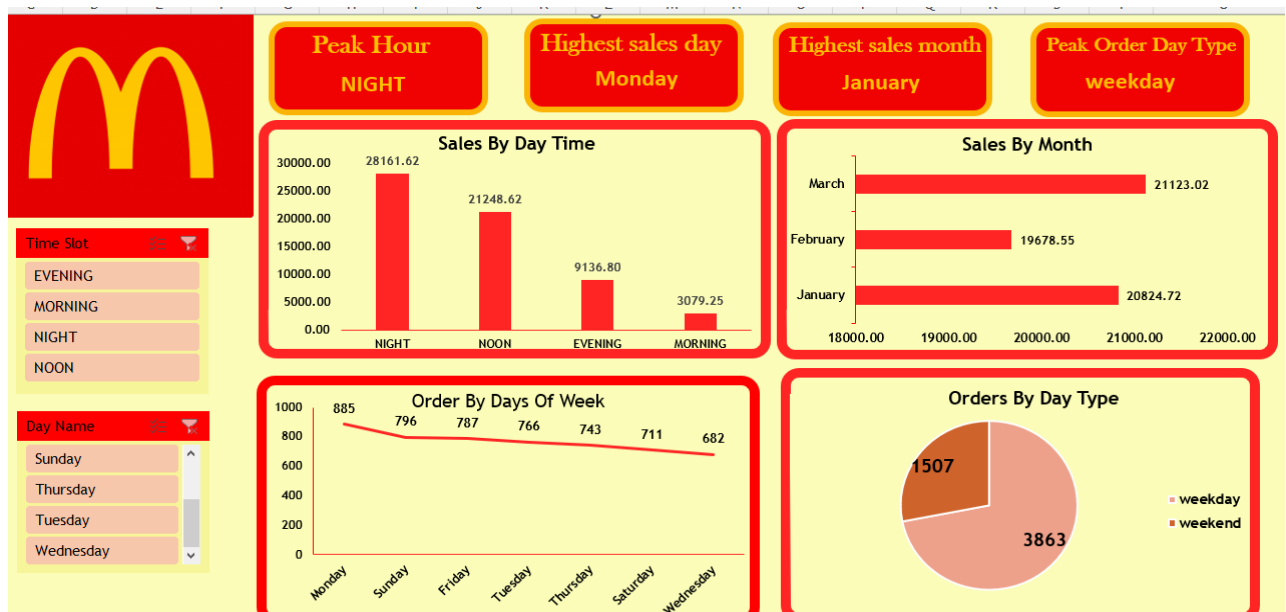
Displays monthly revenue trends.

Orders by Days of Week

Analyzes order frequency across weekdays.

Orders by Day Type

Compares weekday and weekend order counts.



Key Facts from Order Pattern Dashboard

- Night was the peak ordering time slot.
- Monday recorded the highest sales.
- January generated the highest sales month in order patterns.
- Weekdays had significantly higher orders than weekends.
- Customer ordering behavior was strongly time-dependent.

11. Conclusion

The preprocessing phase successfully transformed raw and unstructured transactional data into a clean, reliable, and analysis-ready dataset.

Key preprocessing activities included:

- Data cleaning
- Missing value handling
- Data type standardization
- Feature engineering
- Dataset merging
- Revenue preparation

The transformed dataset enabled accurate:

- Sales analysis
- Revenue tracking
- Product performance analysis
- Time-based order analysis
- Interactive dashboard creation

Overall, preprocessing improved data quality, analytical accuracy, and dashboard performance, creating a strong foundation for business intelligence and decision-making.
